# Dhananjay G. Kharkar

Nagpur, Maharashtra, 441501

+91-9595167618 | dkharkar00@gmail.com | Dhananjay Kharkar | dhananjaykharkar

## SUMMARY

Enthusiastic IoT and AI developer with hands-on experience in environmental monitoring and data-driven decision support. Skilled in ESP32, MQTT, Node-RED, SQL, Firebase, and Streamlit for real-time data acquisition and dashboards. Eager to apply IoT-enabled monitoring, data visualization, and predictive analytics to support environmental research and sustainability studies.

## EDUCATION

**JD College of Engineering & Management** — **Nov 2022 – Pursuing**
*BTech in Artificial Intelligence* ***CGPA: 8.84 / 10*** — *Nagpur, Maharashtra*

**Lokmanya Junior College, Warora** — **June 2021 – April 2022**
*HSC - Maharashtra Board -* ***Percentage - 64%*** — *Warora, Maharashtra*

## PROJECTS

**Real-Time IoT Data Monitoring System** | ESP32, MQTT, Node-RED, SQL Server, Grafana

- Developed and deployed a real-time IoT data pipeline using ESP32 and MQTT to simulate temperature and humidity readings, enabling end-to-end data acquisition and transmission.
- Automated data ingestion and transformation workflows via Node-RED and stored structured sensor data in SQL Server, ensuring consistency and reliability in backend storage.
- Designed dynamic dashboards using Grafana and Node-RED for real-time and historical trend monitoring, and implemented alerting logic to trigger automated Email and SMS notifications on threshold breaches.

**Real-Time Smart Agriculture Dashboard** | ESP32, Firebase, Streamlit, Plotly, OpenWeather API

- Developed a real-time Smart Agriculture Monitoring System using ESP32 to simulate sensor data (temperature, pH, TDS, turbidity) and pushed to Firebase Realtime Database with secure authentication.
- Built an interactive Streamlit dashboard with Plotly-based trend charts, live gauges, threshold-based alerts, and multi-page navigation (Weather, Sensors, Logs).
- Integrated OpenWeatherMap API for real-time weather data and ensured modular architecture for future support with physical sensors.

## INTERNSHIP

**CSE Dept., Visvesvaraya National Institute of Technology** — **June 2025 – July 2025**
*IoT & Data Analytics Intern* — *Nagpur, Maharashtra*

- Worked hands-on with basic sensors including temperature (DHT11), ultrasonic, TDS, GPS, soil moisture, and LDR modules using Arduino UNO and Mega 2560 for foundational data acquisition and control systems.
- Developed a real-time IoT data pipeline using ESP32, MQTT, and Node-RED to simulate and process temperature and humidity readings; stored structured data in SQL Server and visualized trends via Grafana and Node-RED dashboards.
- Implemented threshold-based alerting via Email/SMS and integrated multiple sensor types for real-time monitoring.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Embedded C
**Data Analysis:** Pandas, NumPy, Matplotlib
**Visualization:** Grafana, Streamlit
**Databases:** MySQL, SQL Server, Firebase
**IoT & Embedded Systems:** ESP32, Arduino, MQTT (Mosquitto), Node-RED
**Tools & Frameworks:** VS Code, Git, GitHub

## CERTIFICATIONS

- **The Joy of Computing using Python** – NPTEL
- **Data Analysis with Python** – Coursera
- **Analysing IoT Data using Python** – Datacamp